

- a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(PRICE)$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?

ANS:

```
> red_p <- lm(lprice ~ stormy + mixed + mon + tue + wed + thu, data = dat)
> summary(red_p)
```

Call:

```
lm(formula = lprice ~ stormy + mixed + mon + tue + wed + thu,
    data = dat)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.92281	-0.18819	0.00813	0.22404	0.68381

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-0.35957	0.07915	-4.543	1.50e-05	***
stormy	0.44634	0.07921	5.635	1.51e-07	***
mixed	0.23688	0.07851	3.017	0.00321	**
mon	-0.10825	0.10339	-1.047	0.29753	
tue	-0.06609	0.10103	-0.654	0.51446	
wed	-0.04927	0.10377	-0.475	0.63591	
thu	0.03935	0.10071	0.391	0.69681	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3413 on 104 degrees of freedom

Multiple R-squared: 0.245, Adjusted R-squared: 0.2014

F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05

```
> linearHypothesis(red_p, c("stormy = 0", "mixed = 0"))
```

Linear hypothesis test:

stormy = 0

mixed = 0

Model 1: restricted model

Model 2: lprice ~ stormy + mixed + mon + tue + wed + thu

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

New Reduced-form equation:

$$\text{lprice_hat} = -0.35957 + 0.44634 \cdot \text{stormy} + 0.23688 \cdot \text{mixed} - 0.10825 \cdot \text{mon} - 0.06609 \cdot \text{tue} - 0.04927 \cdot \text{wed} + 0.03935 \cdot \text{thu}$$

The first picture shows that the p-value for mixed variable is 0.00321, which is

smaller than the significance level = 0.05. Therefore, it is statistically significant at the 5% level. The second picture shows that realized F-statistic is 16.14, which is greater than 10, thus we can reject the null hypothesis and infer that the coefficients for stormy and mixed variables are not equal to zero, which means these two variables will influence $\ln(\text{price})$ and satisfies the IV requirement.

- b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.

TABLE 11.5 2SLS Estimates for Fish Demand

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	Prob.
<i>C</i>	8.5059	0.1662	51.1890	0.0000
$\ln(\text{PRICE})$	-1.1194	0.4286	-2.6115	0.0103
<i>MON</i>	-0.0254	0.2148	-0.1183	0.9061
<i>TUE</i>	-0.5308	0.2080	-2.5518	0.0122
<i>WED</i>	-0.5664	0.2128	-2.6620	0.0090
<i>THU</i>	0.1093	0.2088	0.5233	0.6018

ANS:

```
> fit_demand <- ivreg(lquan ~ lprice + mon + tue + wed + thu |
+                     mon + tue + wed + thu + stormy + mixed, data = dat)
> summary(fit_demand)
```

```
Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
      tue + wed + thu + stormy + mixed, data = dat)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.1327 -0.4320  0.0654  0.5133  1.1798
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533 < 2e-16 ***
lprice       -0.93014    0.35340  -2.632  0.00977 **
mon          -0.01190    0.20837  -0.057  0.95456
tue          -0.52583    0.20230  -2.599  0.01068 *
wed          -0.56262    0.20696  -2.719  0.00767 **
thu           0.09987    0.20285   0.492  0.62350
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared: 0.185,    Adjusted R-squared: 0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317
```

From the first picture, as the Coefficients table shows, coefficient estimate for $\ln(\text{price})$, mon , tue , and wed are higher, whereas coefficient for thu is smaller compared to Table 11.5.

c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

ANS:

```
> summary(fit_demand, diagnostics = TRUE)
```

Call:
`ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
 tue + wed + thu + stormy + mixed, data = dat)`

Residuals:

	Min	1Q	Median	3Q	Max
	-2.1327	-0.4320	0.0654	0.5133	1.1798

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	8.54023	0.15661	54.533	< 2e-16	***
lprice	-0.93014	0.35340	-2.632	0.00977	**
mon	-0.01190	0.20837	-0.057	0.95456	
tue	-0.52583	0.20230	-2.599	0.01068	*
wed	-0.56262	0.20696	-2.719	0.00767	**
thu	0.09987	0.20285	0.492	0.62350	

Diagnostic tests:

	df1	df2	statistic	p-value	
Weak instruments	2	104	16.140	7.86e-07	***
Wu-Hausman	1	104	1.489	0.225	
Sargan	1	NA	0.772	0.380	

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
 Multiple R-Squared: 0.185, Adjusted R-squared: 0.1462
 Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317

The result above shows that Sargan Test's p-value is 0.380, which is greater than the significance level 5%. Thus, we cannot reject the null hypothesis H_0 : IVs are not correlated with the error term, which means that the IVs are exogenous and the demand model is stable and robust.

- d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

ANS:

```
> linearHypothesis(red_p, c("mon = 0", "tue = 0", "wed = 0", "thu = 0"))
```

Linear hypothesis test:

```
mon = 0  
tue = 0  
wed = 0  
thu = 0
```

Model 1: restricted model

Model 2: $\ln(\text{price}) \sim \text{stormy} + \text{mixed} + \text{mon} + \text{tue} + \text{wed} + \text{thu}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

The result above shows that the joint test's p-value is 0.6437, which is greater than the significance level 5%. Consequently, we cannot reject the null hypothesis H_0 : The coefficients for mon, tue, wed, and thu are all equal to zero. We can conclude that the price for fish is not influenced by mon, tue, wed, and thu, whereas only weather conditions-stormy and mixed will affect price.

Moreover, since the coefficients for mon, tue, wed, and thu are not significant, when plugging in the fitted value for $\ln(\text{price})$ to the original model, we will find that $\ln(\text{price})_{\text{hat}}$ is just a linear combination of stormy and mixed. (會有共線性問題，造成 supply equation 無法被正確的估計)